# ENZYMES

- Understand the role of enzymes as biological catalysts in metabolic reactions
- Understand the role of digestive enzymes, including the digestion of starch to glucose by amylase and maltase, the digestion of proteins to amino acids by proteases and the digestion of lipids to fatty acids and glycerol by lipases
- Explain the mode of action of enzymes in terms of an active site, enzyme–substrate complex, lowering of activation energy and enzyme specificity, including the lock-and-key hypothesis
- Understand how temperature changes can affect enzyme function, including changes to the shape of the active site.
- Investigate how enzyme activity can be affected by changes in temperature
- Understand how enzyme activity can be affected by changes in enzyme concentration or substrate concentration, as well as changes in pH
- Investigate how enzyme activity can be affected by changes in pH

# Enzymes are Biological Catalysts

## What are catalysts?

A catalyst is a substance that:

- alters or **speeds up** a chemical reaction
- without itself being **chemically** changed at the end of the reaction

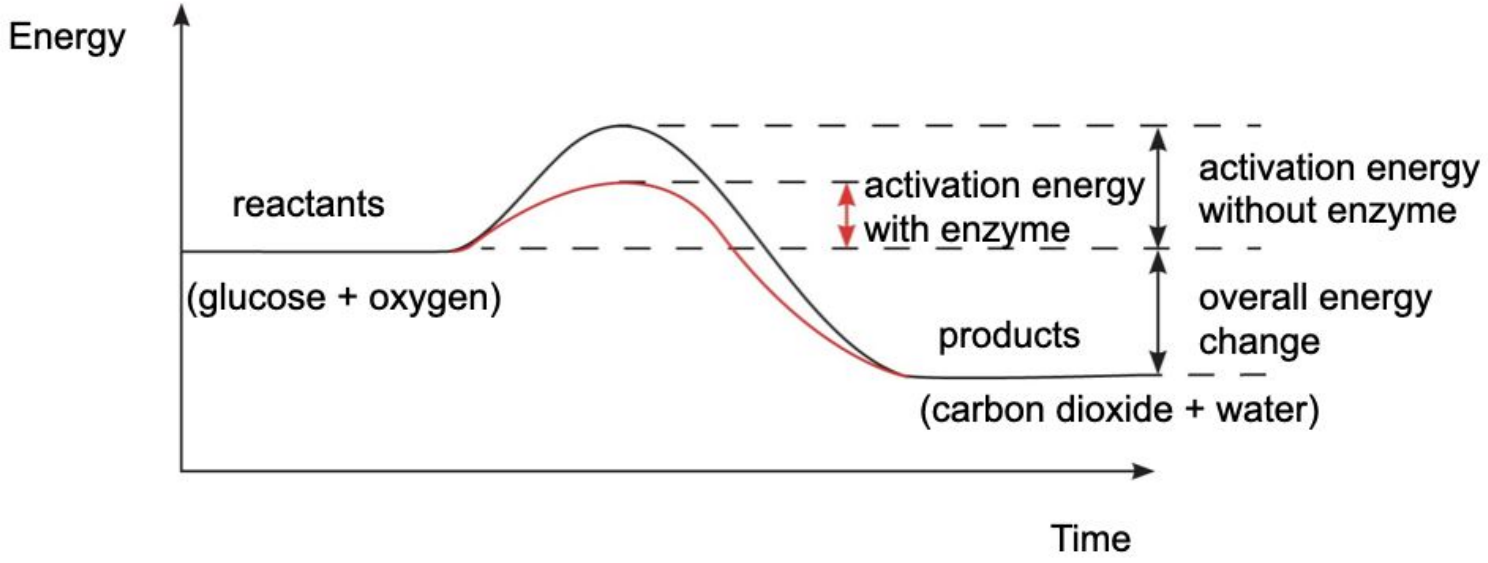
## What are enzymes?

An enzyme:

- is a **protein**
- functions as a **biological** catalyst
- alters or **speeds up** chemical reactions
- remains **chemically unchanged** at the end of the reaction.

# Enzymes lower activation energy

**Activation energy** is the energy needed to start a chemical reaction.



# What reactions do enzymes catalyse?

Digestion

- Some food molecules are large and insoluble in water.
- These molecules cannot diffuse across the **cell membrane**.
- They have to be converted or digested into smaller, simpler molecules in order to do so.
- Examples of digestive enzymes includes:
  a. **Amylase**: Digests starch to maltose
  b. **Maltase**: Digests maltose to glucose
  c. **Protease**: Digests proteins to amino acids
  d. **Lipase**: Digests fats to fatty acids and glycerol

# What reactions do enzymes catalyse?

## Building up of complex substances

- Also known as **anabolic** reactions
- For example, building proteins from amino acids

## Break down of complex substances

- Also known as **catabolic** reactions
- For example, breakdown of glucose during respiration

# Characteristics of Enzymes

- Enzymes **speed up** chemical reactions.
- Enzymes are required in **minute** amounts.
- Enzymes are **highly specific** in action.
    a. **Enzyme specificity**: Each chemical reaction inside a cell is catalysed only by one unique enzyme.
    b. An enzyme is specific due to its three-dimensional shape (**3-D shape**)

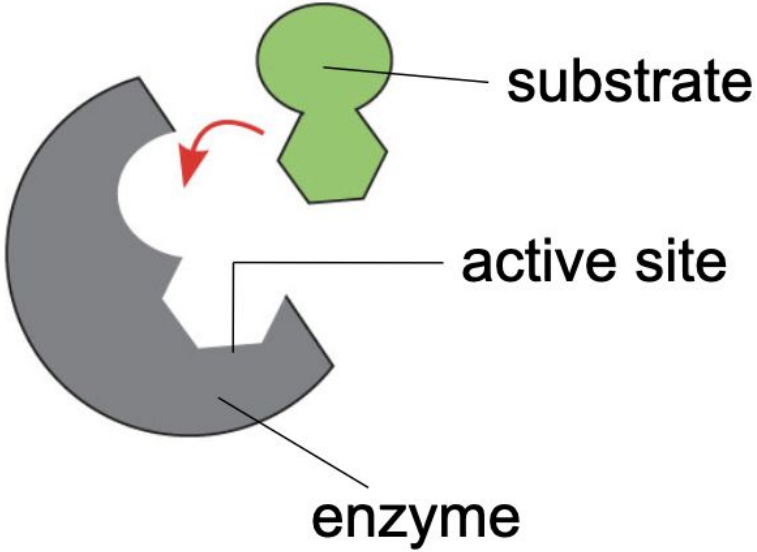
# Enzymes specificity

- The ‘**lock-and-key’** hypothesis explains enzyme specificity.
- According to the ‘lock-and-key’ hypothesis, enzyme reactions depend on the **active site**.

https://www.youtube.com/watch?v=EiMBsgNZh-M

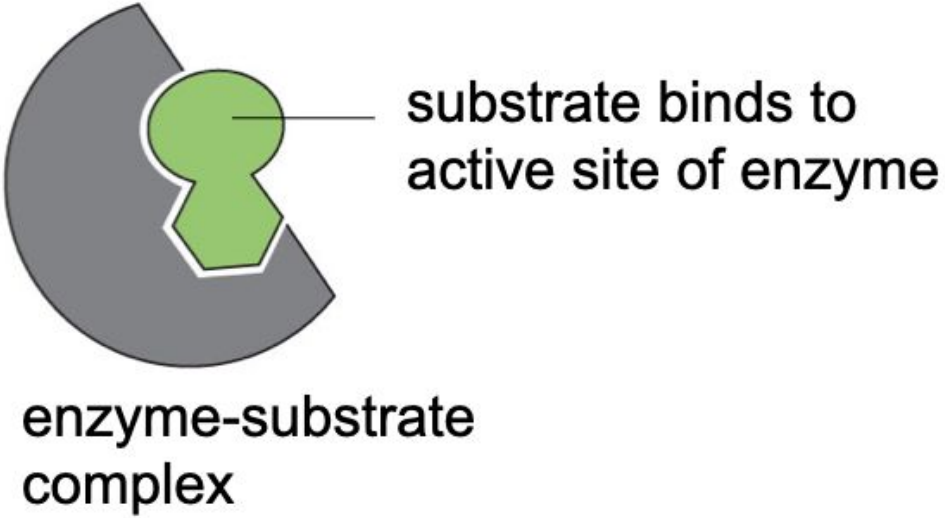
# The ‘lock-and-key’ hypothesis

1. An enzyme has a **specific 3-D shape** which contains an **active site**.



# The ‘lock-and-key’ hypothesis

2. Only the substrate with a 3-D shape **complementary** to that of the active site can fit into the enzyme to form an **enzyme-substrate complex.**



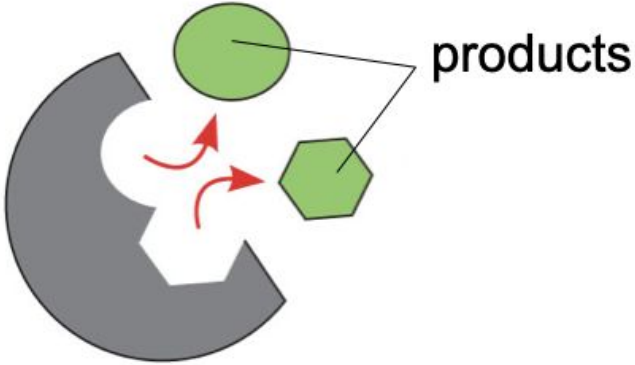
# The ‘lock-and-key’ hypothesis

3. Chemical reaction occurs and the substrate is converted into **products**.

Chemical reaction occurs while substrate is attached to the enzyme

# The ‘lock-and-key’ hypothesis

4. The products then detach from the active site. The enzyme remains **unchanged** at the end of the reaction.



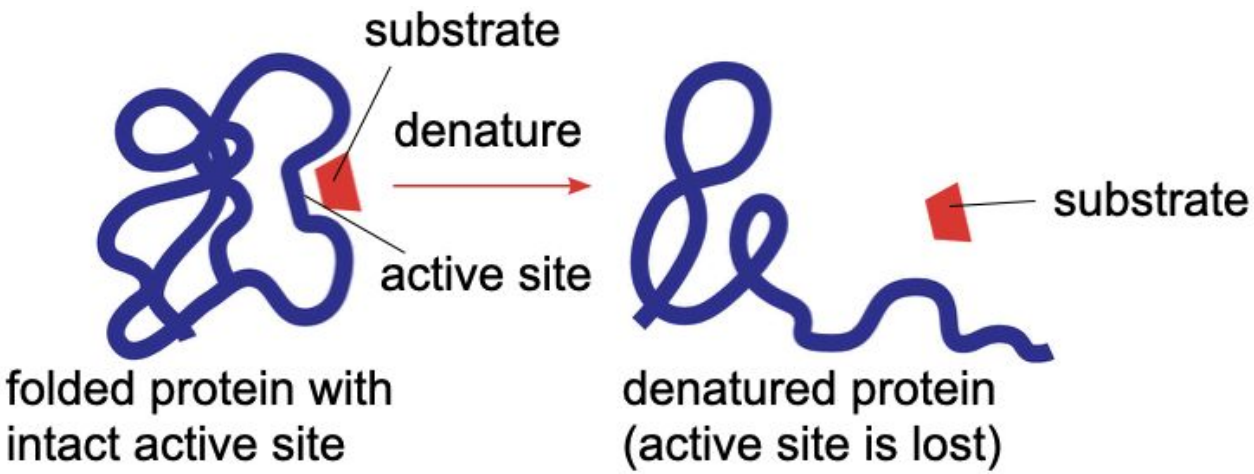

Products leaving active site of enzyme

# Denaturation

- Since the specificity of an enzyme is due to its 3-D shape, any factor that would cause the enzyme to change shape will affect its function.
- The change in 3-D shape of an enzyme or any protein is known as **denaturation**.
- Denaturation can be caused by:
  - **Heat** (extremely high temperatures)
  - Chemicals such as **acids** and **alkalis**

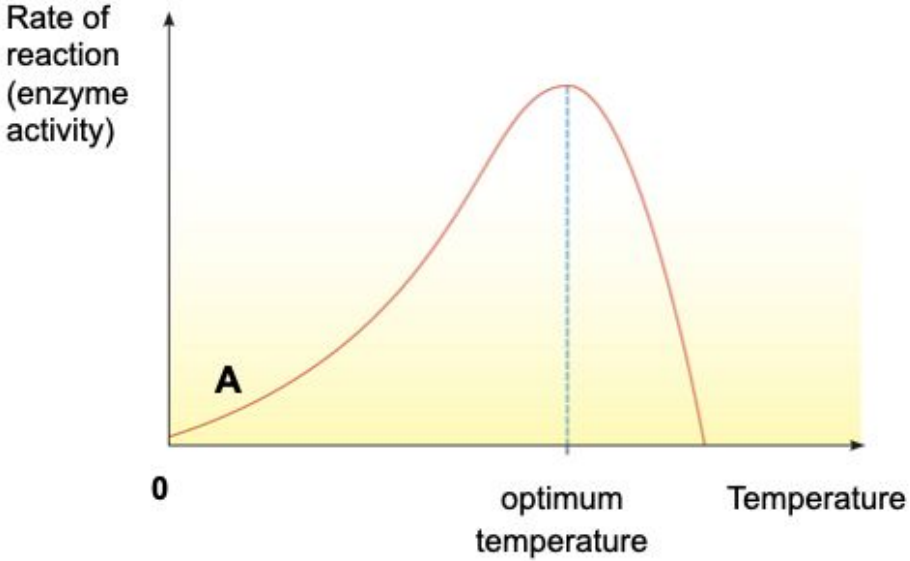
# Denaturation

- When an enzyme is **denatured**, there is loss or change in the active site.
- The substrate can no longer fit into the enzyme's active site.
- Hence, no reaction can take place.



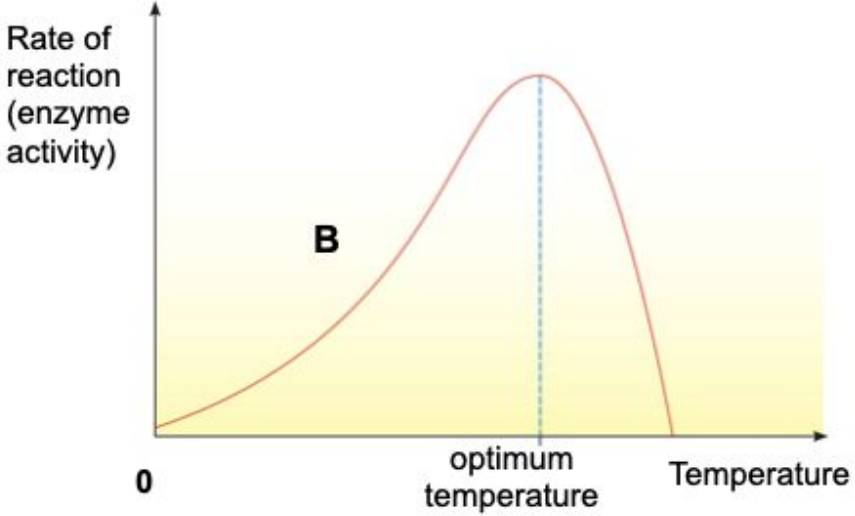
# Enzymes are Affected by Temperature

- Enzymes are **inactive** at very low temperatures.
- The **kinetic energy** is low at low temperatures.
- Hence, chances of substrate molecules colliding with enzymes are **very low.**



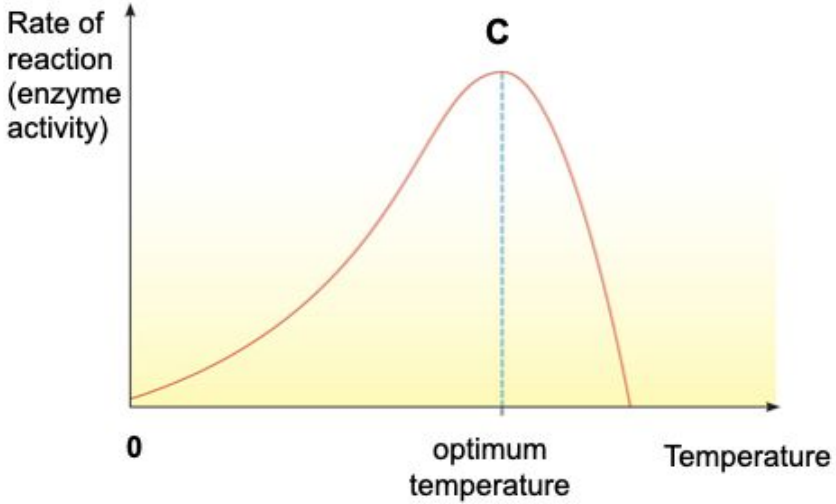
# Enzymes are Affected by Temperature

- As the temperature **rises**, enzyme activity **increases**.
- The increase in the kinetic energy of molecules increases the **chance of collision** between substrate and enzyme molecules.
- This increases the rate of formation of **enzyme-substrate complex.**



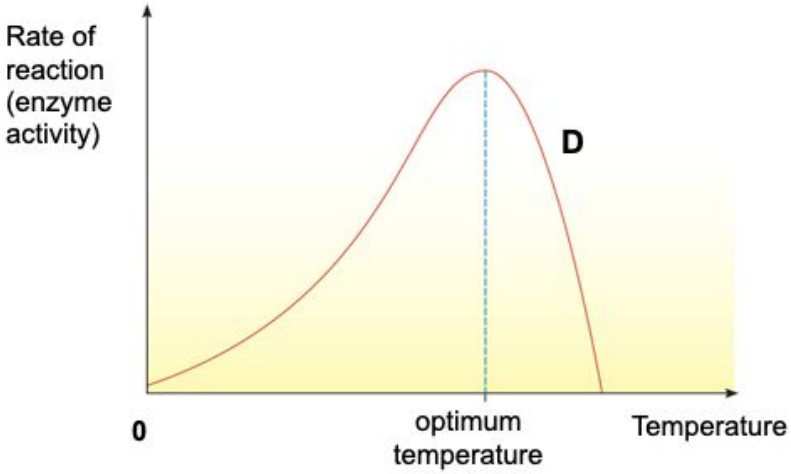
# Enzymes are Affected by Temperature

- The **optimum temperature** is the point when the rate of reaction is highest and the enzyme is most active.
- The optimum temperature for different enzymes varies.
- For most enzymes, the optimum temperature is about **40-45°C**.



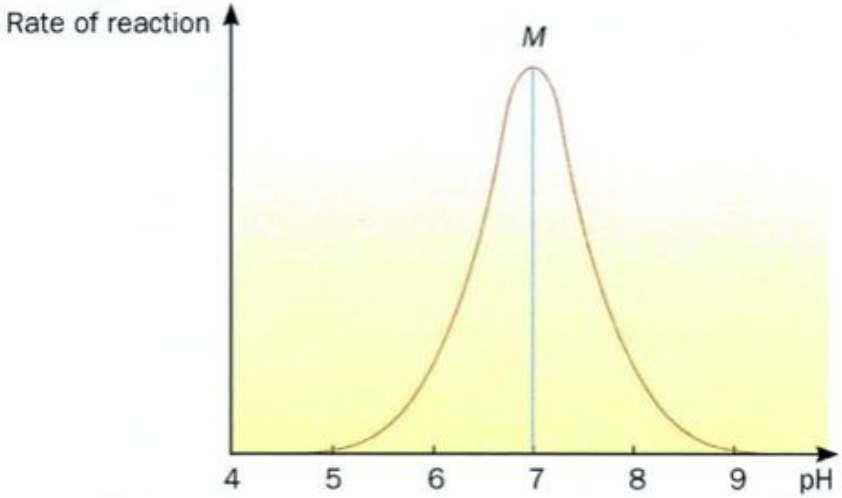
# Enzymes are Affected by Temperature

- Beyond the optimum temperature, the enzyme activity **decreases**.
- High temperature breaks the bond within the enzyme and **changes its 3-D shape**.
- The active site of the enzyme loses its original shape.
- The enzyme is now **denatured**.



# Enzymes are Affected by pH

- Different enzymes have different **optimum pHs**, at which they are most active.
- Extreme changes in pH denature the enzyme and cause it to lose its function.




https://www.youtube.com/watch?v=AFbPHlhI13g

# Enzymes

are

Biological catalysts, which are mainly made of proteins. They speed up the rate of chemical reactions without themselves being chemically changed at the end of the reactions.

## Functions of enzymes

- Digest large food molecules into smaller molecules
- Break down food substances in cells to release energy (cellular respiration)
- Build up or synthesise complex substances
- Break down poisonous substances in cells

## Characteristics of enzymes

- Speed up chemical reactions
- Are required in small amounts
- Are highly specific
- Work best at an optimum temperature and pH
- Catalyse reversible reactions

## Mode of action

- Enzymes provide an alternative pathway with lower activation energy required for a reaction
- Enzymes interact with the substrate according to the 'lock-and-key' hypothesis to form an enzyme–substrate complex

## Class of enzymes

based on the type of reaction catalysed, e.g.

**Hydrolases**
- Carbohydrases
- Proteases
- Lipases

affected by

## Limiting factors

are

Factors that directly affect the rate at which a chemical reaction occurs if their quantity is changed. The value of a limiting factor must be increased in order to increase the rate of the reaction.

e.g.

**Temperature/pH**
- Increase in temperature increases rate of enzyme reaction until optimum temperature is reached.
- Every enzyme has an optimum pH. Increasing or decreasing the pH from the optimum decreases enzyme activity.
- Denaturation of enzyme takes place at extremes of pH and at very high temperatures.